

Martin Birkenbaum

SENIOR SOFTWARE ENGINEER

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Senior Software Engineer with 8+ years of experience designing, building, and deploying resilient, low-latency distributed backend systems and microservices, with a strong focus on scalability, reliability, and security. Proven expertise in real-time systems, data modeling, API design, and full-lifecycle service ownership. Passionate about quality engineering, performance optimization, and driving developer efficiency.

SKILLS

Languages:	Python (Expert), TypeScript, JavaScript, SQL, HTML/CSS
Backend & Distributed Systems:	Microservices, Distributed Systems Design, Scalability, Reliability Engineering, System Architecture, API Design (REST, WebSockets, gRPC), Data Modeling, Real-time Systems, Performance Optimization, Kubernetes, Docker, GitOps, PostgreSQL, NoSQL Databases (e.g., Redis, Cassandra concepts), Caching Strategies, Messaging Queues
Tools & Methodologies:	Git, GitHub Actions, CI/CD, pytest, Test-Driven Development (TDD), Agile Methodologies, Developer Tooling

WORK EXPERIENCE

Senior Software Engineer at **Initech Corporation** 2021-03-01 – 2026-07-01

Led architecture and development of high-throughput microservices and GitOps-driven infrastructure pipelines.

- Designed and implemented high-throughput, Python-based distributed microservices, processing 10M+ daily events with a focus on low-latency and resilience.
- Standardized reliable containerized deployments utilizing Docker, Kubernetes, and GitOps workflows, ensuring scalability and operational excellence.
- Mentored 6 junior/mid-level engineers, establishing robust code styling and quality engineering standards, contributing to maintainable and scalable software.
- Developed custom internal toolchains that reduced CI pipeline execution times by 40%, significantly improving developer efficiency and velocity.

Software Engineer at **Hooli Inc.** 2018-06-01 – 2021-02-01

Backend engineer focused on building RESTful APIs, developer tooling, and automated deployment integrations.

- Developed robust and well-designed REST APIs in Python/Django and TypeScript/Node.js, defining critical data models and service interfaces.
- Integrated third-party APIs and automated webhook processors for external developer platforms, focusing on resilient and secure data exchange.
- Implemented a comprehensive suite of unit and integration tests, raising coverage from 60% to 92%, enhancing system reliability and correctness.

PROJECTS

JobGitOps (Python, Nix, GitHub Actions)

2026-06-01 – 2026-07-01

A GitOps-driven job application and tracking system utilizing GitHub Issues, Projects, and Actions.

- Developed python-jobsby scraper integration and two-pass LLM fit triage
- Designed reproducible local environments via Nix and devenv

EDUCATION

Bachelor of Science in Computer Science at **University of Washington**

2014-09-01 – 2018-06-01

GPA: 3.8